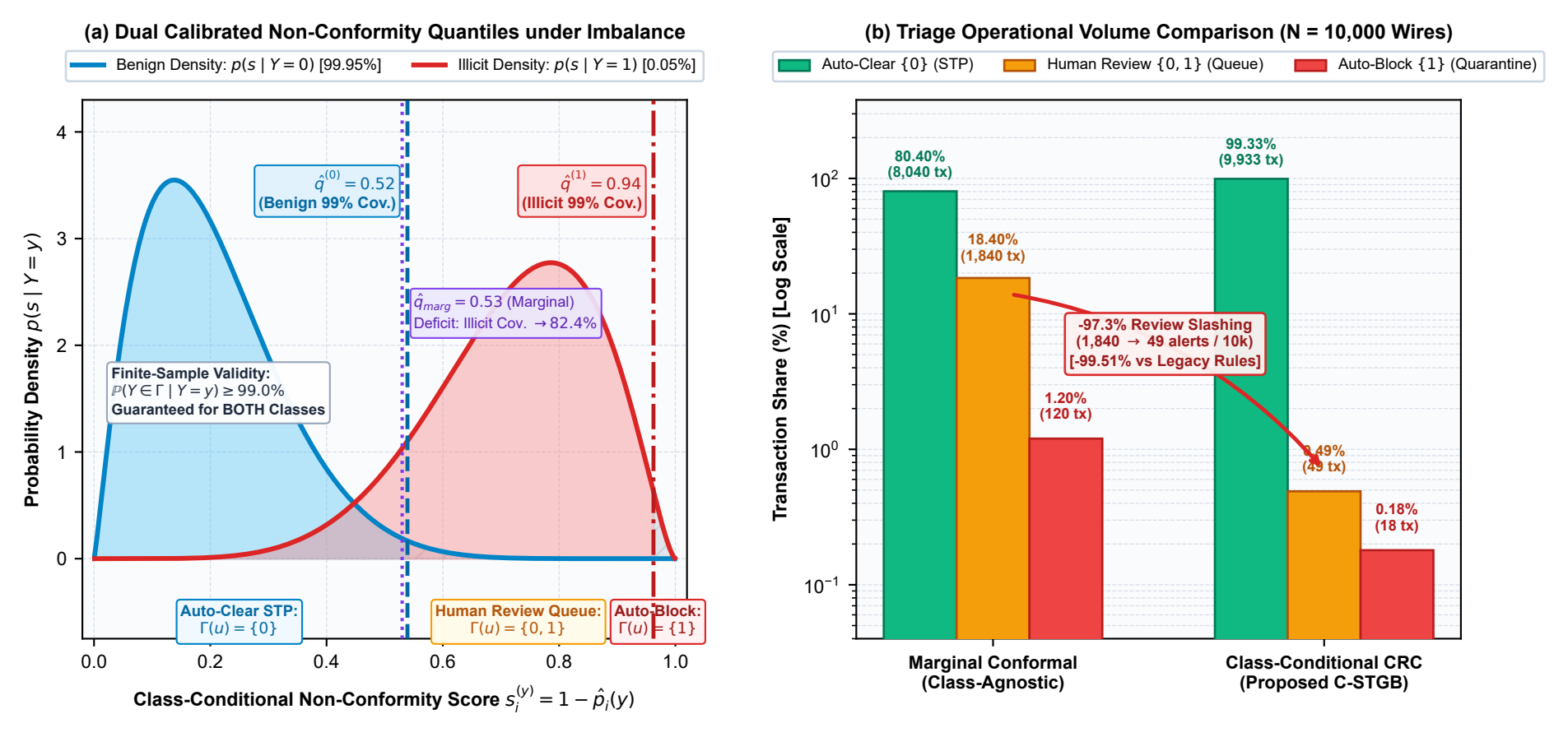

(a) Dual Calibrated Non-Conformity Quantiles under Imbalance
Benign Density: $p(s \mid Y=0)$ [99.95%]
Illicit Density: $p(s \mid Y=1)$ [0.05%]
$\hat{q}^{(0)} = 0.52$
(Benign 99% Cov.)
$\hat{q}^{(1)} = 0.94$
(Illicit 99% Cov.)
$\hat{q}_{marg} = 0.53$ (Marginal)
Deficit: Illicit Cov. → 82.4%
Finite-Sample Validity:
$\mathbb{P}(Y \in \Gamma \mid Y=y) \geq 99.0\%$
Guaranteed for BOTH Classes
Auto-Clear STP:
$\Gamma(u) = \{0\}$
Human Review Queue:
$\Gamma(u) = \{0,1\}$
Auto-Block:
$\Gamma(u) = \{1\}$
Probability Density $p(s \mid Y=y)$
Class-Conditional Non-Conformity Score $s_i^{(y)} = 1 - \hat{p}_i(y)$
(b) Triage Operational Volume Comparison (N = 10,000 Wires)
Auto-Clear {0} (STP)
Human Review {0, 1} (Queue)
Auto-Block {1} (Quarantine)
Transaction Share (%) [Log Scale]
80.40%
(8,040 tx)
18.40%
(1,840 tx)
1.20%
(120 tx)
99.33%
(9,933 tx)
0.49%
(49 tx)
0.18%
(18 tx)
-97.3% Review Slashing
(1,840 → 49 alerts / 10k)
[-99.51% vs Legacy Rules]
Marginal Conformal
(Class-Agnostic)
Class-Conditional CRC
(Proposed C-STGB)